

i Front page TIØ4145 2025
Examination paper for TIØ4145 Corporate Finance
Date: 11.12.2025
Time: 15.00-19.00

Course contact: Maria Lavrutich
Present at the exam location: No

Permitted examination support material: D - No printed or hand-written support material is allowed. A specific basic calculator is allowed.

OTHER INFORMATION

Read the questions carefully and make your own assumptions. In your answers, explain clearly what assumptions you have made and how you have understood or limited the assignment.

If there are direct errors or omissions in the assignment set and you cannot make your own assumptions, please refer to the information about complaints regarding formal errors on the NTNU website "Explanation of grades and appeals".

SPECIFIC INFORMATION FOR YOUR COURSE

Paper drawings: All questions should be answered directly in Inspera. However, for sections 2,3, and 4 you may choose to answer on handwritten sheets. At the bottom of the question, you will find a seven-digit code. Fill in this code in the top left corner of the sheets you wish to submit.

Please note that only blue or black ballpoint pens are allowed

We recommend that you do this during the exam. If you require access to the codes after the examination time ends, click "Show submission".

You are responsible for filling in the correct codes on the handwritten sheets. Therefore, read the cover sheet carefully. The Examination Office cannot guarantee that incorrectly completed sheets will be added to your assignment.

Weighting: Multiple choice questions in Section 1 are graded as follows: 2 points are given for correct answer; 1 point is subtracted for wrong answer; 0 points is given when the question is left unanswered. Only one answer is correct.

Withdrawing from the exam:

If you wish to submit a blank test/withdraw from the exam for another reason, go to the menu in the top right-hand corner and click "Submit blank". This cannot be undone, even if the test is still open.

Access to your answers:

After the exam, you can find your answers under previous tests in Inspera. Be aware that it may take a working day until any hand-written material is available in "previous tests".

1.1 Beta

What does a *beta of a stock relative to the market* measure?

Select one alternative:

- ☐ Total risk
- ☐ Diversifiable risk
- ☐ Overall volatility of the financial markets
- ☐ Systematic risk

Maximum marks: 2

1.2 Private Equity Funds

What differentiates private equity funds from other types of funds?

Select one alternative:

- ☐ They primarily focus on short-term trading in public equity markets
- ☐ They only use debt in their investments
- ☐ They only rely on Leveraged Buyout (LBO) models as their valuation methodology
- ☐ They invest in unlisted companies

Maximum marks: 2

1.3 Risk-Neutral Probabilities

TunnelVision AS has 12 million shares outstanding with a current share price of \$40 per share. In one year, the share price is equally likely to be \$60 or \$30. The risk-free interest rate is 5%. What is the risk-neutral probability that TunnelVision AS's stock price will increase?

Select one alternative:

- ☐ 0.25
- ☐ 0.1
- ☐ 0.6
- ☐ 0.4

Maximum marks: 2

1.4 Payout policy

What is a firm's payout policy?

Select one alternative:

- ☐ The way a firm chooses between alternative methods of distributing free cash flow to its equity holders.
- ☐ The rules governing how a firm must disclose information to shareholders.
- ☐ The decision of whether the firm should invest in new projects rather than keep cash internally.
- ☐ The method a firm uses to determine its capital structure.

Maximum marks: 2

1.5 Solvency II

Which of the following statements about Solvency II is FALSE?

Select one alternative:

- ☐ Solvency II has a risk-based approach that enables to assess the “overall solvency” of insurance and reinsurance undertakings through quantitative and qualitative measures
- ☐ Solvency II is a regulatory framework that applies to all financial institutions in the EU, including banks, investment firms, and insurance companies.
- ☐ Solvency II is the prudential regime for insurance and reinsurance undertakings in the EU.
- ☐ Solvency II sets out requirements to ensure the adequate protection of policyholders and beneficiaries.

Maximum marks: 2

1.6 Option pricing

In most securities (e.g., stocks, corporate bonds), we estimate a required return (for example using CAPM) and discount expected cash flows at that rate. Why do we not typically price options by taking the expected payoff and discounting it at a CAPM-based discount rate?

Select one alternative:

- ☐ Because option payoffs depend on the underlying asset's price, any risk is already fully reflected in the stock's required return, so the option itself has no additional risk and should be discounted at the risk-free rate.
- ☐ Because CAPM assumes a linear relation between returns and the market, it cannot be used for any derivative or contingent claim.
- ☐ Because an option's payoff is a nonlinear function of the underlying asset, its risk and beta change across states and over time, so a single constant required return for the option is not consistent with its risk profile.
- ☐ Because option prices are very volatile, so it is impossible to compute any expected return for them.

Maximum marks: 2

1.7 CAPM

Which of the following statements is FALSE?

Select one alternative:

- ☐ A conclusion of the CAPM is that investors should hold the market portfolio even if they do not have high trading skills.
- ☐ The CAPM assumption of homogeneous expectations is not necessarily a good description of the real world.
- ☐ Even naive investors with no information should hold the market portfolio.
- ☐ A conclusion of the CAPM is that investors should hold the market portfolio only if they have high quality information.

Maximum marks: 2

1.8 Index Investing

Which statement best describes the key difference between mutual funds (including index funds) and exchange traded funds (ETFs)?

Select one alternative:

- ☐ Unlike mutual funds, an ETF trades like a common stock on a stock exchange.
- ☐ Mutual funds always track a benchmark index, while ETFs can only be actively managed.
- ☐ Mutual funds and ETFs both trade continuously on stock exchanges, but ETFs charge higher fees because they track more complex indices.
- ☐ ETFs represent ownership in only one stock at a time, whereas mutual funds must hold a diversified portfolio.

Maximum marks: 2

1.9 Share repurchases

Under what condition is a share repurchase a good use of cash for the shareholders who continue to hold their shares after the repurchase?

Select one alternative:

- ☐ It is always beneficial, since all shareholders gain equally from a repurchase, regardless of whether the stock is over- or undervalued.
- ☐ When management believes the stock is undervalued.
- ☐ When the firm wants to increase the number of shares outstanding.
- ☐ When managers want to signal confidence by distributing cash, even if they think the stock is overvalued.

Maximum marks: 2

1.10 Default and Bankruptcy in a Perfect Market

Which of the following statements is FALSE?

Select one alternative:

- ☐ Economic distress is a significant decline in the value of a firm's assets, whether or not it experiences financial distress due to leverage.
- ☐ An important consequence of leverage is the risk of bankruptcy.
- ☐ In a perfect capital market, Modigliani–Miller capital-structure irrelevance still holds even when debt is risky and the firm may default
- ☐ Whether default occurs depends on the cash flows, not on the relative values of the firm's assets and liabilities.

Maximum marks: 2

2.1 Question 2.1 (15%)

Assume there are four default-free bonds with the following prices and future cash flows:

Bond	Price Today	Cash flow Year 1	Cash flow Year 2	Cash flow Year 3
A	934.58	1000	0	0
B	881.66	0	1000	0
C	1118.21	100	100	1100
D	839.62	0	0	1000

- i) Do these bonds present an arbitrage opportunity?
- ii) If so, how would you take advantage of this opportunity? If yes, describe the strategy and find the arbitrage profits. If not, why not?

Fill in your answer here

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Maximum marks: 15

2.2 Question 2.2 (10%)

Disregard Question 2.1. Consider now another company Scoops Ahoy Corp. that has issued zero-coupon corporate bonds with a four-year maturity and a face value of \$1000.

Investors believe there is a 30% chance that Scoops Ahoy will default on these bonds by maturity. If the firm defaults, investors expect to receive only 40 cents per dollar of face value they are owed.

Investors require a 7% expected return on their investment in these bonds.

- i) What is the current price of Scoops Ahoy Corp.'s bond?
- ii) What is its yield to maturity (YTM), based on the promised payoff at maturity?
- iii) In this setting, does the expected return equal the YTM? Explain, why or why not?

Fill in your answer here

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Maximum marks: 10

3.1 Question 3.1 (4%)

NotSoGreen Inc. is a global manufacturer of high-quality steel for the automotive and construction sectors. The company is under increasing scrutiny for several environmental and social impacts, including high greenhouse gas emissions, eutrophying discharges affecting nearby freshwater ecosystems, and evidence of forced labor in parts of its upstream supply chain.

The company's current sustainability impacts are summarized as follows:

- Greenhouse gas emissions: 300 000 kg CO₂ equivalents (CO₂-eq) per year
- Eutrophying emissions: 1500 kg phosphorus equivalents (P-eq) per year
- Forced labor: 0.5 full-time equivalent (FTE) in the supply chain

To address these issues, the firm has established the following sustainability targets:

- Reduce CO₂ emissions by at least 25% ($\leq 225\,000$ kg CO₂-eq/year)
- Reduce eutrophying emissions by at least 50% (≤ 750 kg P-eq/year)
- Eliminate forced labor

The firm must now decide which sustainability project to prioritize as it adapts to international ESG standards. More specifically, it considers investing in one of the following projects.

	Project A: Emissions Reduction Initiative	Project B: Ethical Supply Chain Program	Project C: Circular Economy & Recycling
Focus	• Upgrade of production equipment	• Restructuring the supply chain, audits, and community training	• Recycling steel scrap to reduce raw-material inputs
Sustainability impact	• CO₂ reduced by 100 000 kg • Eutrophying emissions reduced by 750 kg	• Eliminates all 0.5 FTE of forced labor	• CO₂ reduced by 50 000 kg • Eutrophying emissions reduced by 375 kg
Economic impact	• Investment: €60 million • Cash flow: €20 million/year	• Investment: €70 million • Cash flow: €25 million/year	• Investment: €50 million • Cash flow: €22 million/year

Assume that the investment is made today, while the listed annual cash flows and sustainability impacts occur once per year for the next five years, starting one year from now. Assume further that the financial cashflows are discounted at 5% per year, while environmental and social impacts are discounted at 2% per year.

Compare the three projects using **constrained NPV method**, incorporating both their financial value and the extent to which each project meets NotSoGreen Inc.'s sustainability targets. Based on this combined assessment, which project if any should be prioritized?

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3.2 Question 3.2 (10%)

Keeping all previous assumptions from Question 3.1. Evaluate the projects using an **integrated NPV approach** that incorporates monetized sustainability impacts.

Use the following monetization factors from for each impact category:

- Greenhouse gas emissions: € 0.163 per kg CO₂-eq
- Eutrophying emissions: € 209 per kg P-eq
- Forced labor: €123 000 per FTE

Using these monetization factors together with the financial and sustainability data provided earlier, calculate the integrated NPV for each project.

1. Based on this, which project if any should NotSoGreen Inc. prioritize?
2. Would different strategic priorities lead the company to choose a different project than the one with the highest integrated NPV? Explain.

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3.3 Question 3.3 (6%)

Sustainability assessments that integrate financial, environmental, and social impacts often face several methodological and practical difficulties.

Describe two major challenges associated with conducting such integrated sustainability assessments. Discuss why these challenges occur and how they affect the reliability or interpretation of the results.

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Maximum marks: 6

Maximum marks: 15

4.2 Question 4.2 (10%)

Keep all assumptions from Questions 3.1.

- ii) Determine the value of AlmostFinished Infrastructure Inc. as an unlevered company, ignoring the effects of debt.
- iii) Calculate the value of AlmostFinished Infrastructure Inc.'s levered value and the value of its interest tax shield.

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4.3 Question 4.3 (10%)

Keep all assumptions from Questions 3.1 and 3.2, apart from the assumption that taxes are the only market imperfection. Instead, assume that the management is increasingly concerned that the firm's high leverage ($D/E = 0.70$) may expose AlmostFinished Infrastructure Inc. to significant financial distress. They have now ordered an external analysis to evaluate its potential impact. The consultants provide the following estimates of the expected present value of bankruptcy costs at different leverage levels as follows:

Debt-to-equity ratio	Expected present value of bankruptcy costs
0.2	0
0.4	5 mil NOK
0.7	50 mil NOK

i) Is the management right to be concerned that the current debt-to-equity ratio of 0.7 is too high? Among the three debt-to-equity ratios in the table above, which capital structure yields the highest firm value?

ii) When managers of AlmostFinished Infrastructure worry about "bankruptcy costs," what kinds of real-world costs do they have in mind? Give 3 concrete examples of such costs.

Fill in your answer here

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